SkillBridge | Git Essentials

Track your first change

Create a small repository and understand the working tree, staging area and commit history.

Git records snapshots of a project. The working tree contains the files you edit. The staging area selects changes for the next snapshot. A commit records that snapshot with a message and a link to the previous commit.

Practise in a new folder rather than an existing assignment. Run: mkdir git-practice, then cd git-practice, then git init. Create a file named notes.txt containing one sentence. Run git status to see that the new file is untracked.

Run git add notes.txt to stage the file. Run git diff --cached to inspect the staged change. Commit it with git commit -m "Add practice notes". If Git asks for an identity, configure your own name and email before retrying.

Edit the sentence and run git diff. This compares the working file with the staged version. Stage the update, review git diff --cached again and make a second focused commit. Run git log --oneline to inspect the two snapshots.

Checkpoint: explain why saving a file does not automatically create a Git commit. Your answer should distinguish editing, staging and committing. Never stage passwords, database credentials or build output.

Work on a feature branch

Use a branch to isolate changes and review them before combining work.

A branch is a movable name pointing to a commit. It lets you develop a change without immediately moving the main branch. It is not a separate backup of every file.

In the practice repository, run git switch -c improve-notes. Add a second paragraph to notes.txt, stage it, inspect the staged diff and commit with a descriptive message.

Run git switch main if your initial branch is named main; otherwise use the branch name shown by git branch. Run git merge improve-notes to combine the completed practice change. If no competing commits exist, Git may fast-forward the branch pointer.

A merge conflict means Git could not safely combine overlapping changes. Open the reported files, decide the intended final text, remove conflict markers, test the result, stage the resolved files and complete the merge. Do not choose a version blindly.

For group work, get the latest shared branch before starting, use a feature branch, commit source files rather than a ZIP, and review the changes before merging. Use git status to check for uncommitted work before switching tasks.

Checkpoint: describe which branch you would use for a quiz-editor change and what you would review before integrating it.

Use a disposable practice folder/database. Original instructional material prepared for SkillBridge.